

A Consolidated Formula Atlas for the Thue–Morse Sequence

Digital, valiative, Pascal, automatic, generating, Prouhet, Fourier, spectral, and analytic representations

Editorial consolidation and expansion of four research-frontier drafts

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Abstract

Four substantial drafts in the non-formalized research frontier of the ProveIt Fabius-function development independently assembled exact formulae for the Thue–Morse sequence. Their mathematical cores overlap, but each contains material absent from the others. This article merges the four into one notation and one proof architecture. It retains the distinct digit, floor, valuation, carry, Pascal-row, multinomial, automatic, generating, finite-difference, Fourier, spectral, algebraic, infinite-product, and Fabius-function formulae; removes duplicate derivations; and makes explicit the maps between them.

The unifying object is the finite product

$$P_m(z) = \prod_{j=0}^{m-1} (1 - z^{2^j}) = \sum_{n=0}^{2^m-1} \varepsilon_n z^n, \quad \varepsilon_n = (-1)^{s_2(n)}.$$

Its coefficients encode binary parity, its zero of order m at $z = 1$ encodes Prouhet cancellation, its values on the unit circle are Fourier sums, and its normalized squared modulus yields the Riesz products of the Thue–Morse diffraction measure. Around this backbone we derive arithmetic formulae based on central binomial and Catalan valuations, carry counts, Lucas support, log-free Pascal statistics, Bell polynomials, determinant formulae, exact cubature, sparse-submask identities, autocorrelation, and characteristic-two algebraicity.

New material added in the consolidation includes exact closed formulae for signed residue-class sums modulo 3, a functional equation for the one-sided autocorrelation generating series, a sharp discussion of the Gelfond exponent and cubic-root extremizers, and a compact synthesis of transcendence, irrationality-measure, overlap-free, Hankel-determinant, and spectral results from the literature. A final section identifies statements that are especially well suited for formalization in Lean.

Provenance and priority. This is an editorial consolidation of four exploratory drafts, not a claim of historical priority for classical identities. Formulae proved directly below are separated from literature theorems by explicit attribution. The source comparison is with the four files named `Further_Formulae_for_the_Thue_Morse_Sequence-1.tex` through `-4.tex` in `Analysis/FabiusFunction/docs/non-formalized-research-frontiers/drafts`.

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1 Scope, normalization, and the master dictionary

1.1 Two normalizations

Write the binary expansion of $n \geq 0$ as

$$n = \sum_{j \geq 0} b_j(n) 2^j, \quad b_j(n) \in \{0, 1\},$$

with only finitely many nonzero digits. Its binary digit sum is

$$s_2(n) = \sum_{j \geq 0} b_j(n).$$

The zero–one and signed Thue–Morse sequences are

$$\tau(n) = s_2(n) \bmod 2 \in \{0, 1\}, \quad \varepsilon_n = (-1)^{s_2(n)} = 1 - 2\tau(n) \in \{1, -1\}. \quad (1)$$

Thus

$$\tau(n) = \frac{1 - \varepsilon_n}{2}, \quad \varepsilon_n = 1 - 2\tau(n). \quad (2)$$

We use P_m for the length- 2^m signed generating polynomial, E for its infinite analytic limit in the unit disc, and T for the zero–one generating function:

$$P_m(z) = \sum_{n=0}^{2^m-1} \varepsilon_n z^n, \quad E(z) = \sum_{n \geq 0} \varepsilon_n z^n, \quad T(z) = \sum_{n \geq 0} \tau(n) z^n. \quad (3)$$

1.2 The master formula atlas

The following display is a navigation map rather than a single theorem. Every line is proved, refined, or situated later.

The individual value.

$$\begin{aligned}\varepsilon_n &= (-1)^{s_2(n)} = \prod_{j \geq 0} (1 - 2b_j(n)) = (-1)^{n - \sum_{j \geq 1} \lfloor n/2^j \rfloor} \\ &= (-1)^{v_2(n!)} (-1)^n = (-1)^{v_2\binom{2n}{n}} = \prod_{j \geq 0} \cos(\pi \lfloor n/2^j \rfloor), \\ \tau(n) &= \frac{1 - \varepsilon_n}{2} = \left[v_2\binom{2n}{n} \right]_2 = [s_2(n)]_2.\end{aligned}$$

Dyadic recursion and blocks.

$$\varepsilon_{2n} = \varepsilon_n, \quad \varepsilon_{2n+1} = -\varepsilon_n, \quad (\varepsilon_{2^m}, \dots, \varepsilon_{2^{m+1}-1}) = -(\varepsilon_0, \dots, \varepsilon_{2^m-1}).$$

Pascal and carries.

$$\begin{aligned}\#\{k : \binom{n}{k} \text{ odd}\} &= 2^{s_2(n)}, \quad v_2\binom{a+b}{a} = s_2(a) + s_2(b) - s_2(a+b), \\ \varepsilon_{a+b} &= \varepsilon_a \varepsilon_b (-1)^{v_2\binom{a+b}{a}}.\end{aligned}$$

Finite product, Prouhet, and Fourier.

$$\begin{aligned}P_m(z) &= \prod_{j=0}^{m-1} (1 - z^{2^j}), \quad P_m(z) = (1 - z)P_{m-1}(z^2), \quad \text{ord}_{z=1} P_m = m, \\ \sum_{n < 2^m} \varepsilon_n n^r &= 0 \quad (r < m), \quad \widehat{\varepsilon}_m(k) = P_m(e^{-2\pi i k/2^m}).\end{aligned}$$

Infinite product and Mahler equations.

$$\begin{aligned}E(z) &= \prod_{j \geq 0} (1 - z^{2^j}), \quad E(z) = (1 - z)E(z^2), \\ T(z) &= \frac{1}{2(1 - z)} - \frac{1}{2}E(z), \quad T(z) = (1 - z)T(z^2) + \frac{z}{1 - z^2}.\end{aligned}$$

1.3 Four complementary viewpoints

The four drafts approach the same sequence from different directions. Their most useful synthesis is the following four-way dictionary.

Viewpoint	Basic object	Characteristic operation
Digital	binary digits $b_j(n)$ and digit sum $s_2(n)$	append a digit; split into even and odd indices
Valuative/Pascal	$v_2(n!)$, $v_2\binom{a+b}{a}$, odd entries in row n	count carries; use Legendre, Kummer, and Lucas
Generating/Prouhet	$P_m(z) = \prod_{j < m} (1 - z^{2^j})$	factor at $z = 1$; differentiate; extract coefficients

Viewpoint	Basic object	Characteristic operation
Harmonic/ spectral	$P_m(e^{2\pi i x})$ and normalized Riesz products	evaluate on roots of unity; square the modulus; pass to a weak limit

The rest of the article repeatedly moves around this dictionary. A proof is usually shortest in one column and most informative in another.

2 Binary recursion, morphisms, blocks, and automata

2.1 The fundamental recursion

Theorem 2.1 (Dyadic recursion). *For every $n \geq 0$,*

$$\tau(2n) = \tau(n), \quad \tau(2n+1) = 1 - \tau(n), \quad (4)$$

and equivalently

$$\varepsilon_{2n} = \varepsilon_n, \quad \varepsilon_{2n+1} = -\varepsilon_n. \quad (5)$$

Proof. Multiplication by 2 shifts every binary digit one place to the left and does not change the digit sum. Adding 1 after that inserts a final 1, so $s_2(2n) = s_2(n)$ and $s_2(2n+1) = s_2(n) + 1$. Reduce modulo 2 or exponentiate -1 . $\square$

Starting from $\tau(0) = 0$, the recursion is equivalent to the fixed point of the length-two morphism

$$\mu : 0 \mapsto 01, \quad 1 \mapsto 10. \quad (6)$$

For signed symbols it is the substitution $+\mapsto+-$ and $-\mapsto-+$. Thus the words

$$W_m = (\varepsilon_0, \varepsilon_1, \dots, \varepsilon_{2^m-1})$$

satisfy

$$W_{m+1} = W_m(-W_m). \quad (7)$$

The first six signed blocks are

$$\begin{aligned} W_0 &= (+), \\ W_1 &= (+, -), \\ W_2 &= (+, -, -, +), \\ W_3 &= (+, -, -, +, -, +, +, -), \\ W_4 &= (+, -, -, +, -, +, +, -, -, +, +, -, +, -, -, +). \end{aligned}$$

Corollary 2.2 (Complementation on dyadic blocks). *For $0 \leq r < 2^m$,*

$$\varepsilon_{2^m+r} = -\varepsilon_r, \quad \tau(2^m+r) = 1 - \tau(r). \quad (8)$$

Proof. The binary expansion of $2^m + r$ has a new leading 1 followed by the m -digit expansion of r . $\square$

2.2 Concatenation and digit disjointness

Proposition 2.3 (No-carry concatenation). *If $0 \leq r < 2^m$, then for every $q \geq 0$,*

$$s_2(2^m q + r) = s_2(q) + s_2(r), \quad \varepsilon_{2^m q + r} = \varepsilon_q \varepsilon_r. \quad (9)$$

More generally, if $a \& b = 0$ bitwise, then $\varepsilon_{a+b} = \varepsilon_a \varepsilon_b$.

Proof. The binary digits of $2^m q$ occupy positions $m, m+1, \dots$, whereas those of r occupy positions below m . Hence addition produces no carry and the digit sums add. The second statement is the same argument without choosing a block boundary. $\square$

Bitwise exclusive-or gives a carry-free law without any disjointness hypothesis. Since each bit of $a \oplus b$ is the sum modulo two of the corresponding bits of a and b ,

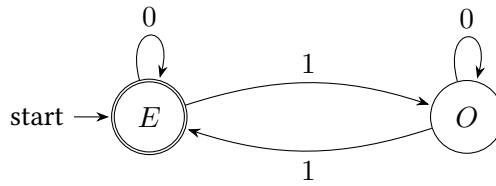
$$\boxed{\varepsilon_{a \oplus b} = \varepsilon_a \varepsilon_b.} \quad (10)$$

Thus ε is a character of the Boolean group of finite binary strings.

Remark 2.4. The correction to [equation \(9\)](#) when ordinary addition creates carries is exactly the Kummer factor in [theorem 4.5](#). Thus the simplest digital formula and the valuation formula are not separate facts: Kummer’s theorem measures the failure of digit-disjoint multiplicativity.

2.3 The two-state automaton and the 2-kernel

Reading the binary expansion of n from most significant digit to least, the state records the parity of the number of 1’s seen so far.



Here E outputs $\tau = 0$ and $\varepsilon = +1$, while O outputs $\tau = 1$ and $\varepsilon = -1$.

Definition 2.5. The 2-kernel of a sequence $a = (a_n)_{n \geq 0}$ is

$$\mathcal{K}_2(a) = \{(a_{2^e n + r})_{n \geq 0} : e \geq 0, 0 \leq r < 2^e\}.$$

Proposition 2.6 (Minimal kernel). *The signed Thue–Morse sequence has*

$$\mathcal{K}_2(\varepsilon) = \{\varepsilon, -\varepsilon\},$$

and the zero–one sequence has

$$\mathcal{K}_2(\tau) = \{\tau, 1 - \tau\}.$$

Consequently both are 2-automatic.

Proof. Iterating [equation \(9\)](#) gives $\varepsilon_{2^e n + r} = \varepsilon_n \varepsilon_r$. Since $\varepsilon_r \in \{\pm 1\}$, each kernel element is either ε or $-\varepsilon$, and both occur. The zero–one statement follows from [equation \(2\)](#). $\square$

2.4 Tensor and Walsh descriptions

Let $n = (b_{m-1} \cdots b_1 b_0)_2$ with leading zeroes allowed. Then

$$\varepsilon_n = \prod_{j=0}^{m-1} (-1)^{b_j(n)} = \bigotimes_{j=0}^{m-1} (1, -1)_{b_j(n)}. \quad (11)$$

Hence the length- 2^m vector is the Kronecker power

$$(\varepsilon_0, \dots, \varepsilon_{2^m-1}) = (1, -1)^{\otimes m}. \quad (12)$$

In the additive group $(\mathbb{F}_2)^m$, it is the Walsh character

$$\varepsilon_n = (-1)^{\langle \mathbf{1}, \mathbf{b}(n) \rangle}, \quad \mathbf{1} = (1, \dots, 1). \quad (13)$$

This viewpoint makes orthogonality immediate: over the full Boolean cube, ε is orthogonal to every Walsh character except itself.

2.5 Overlap-free structure

A finite or infinite word contains an *overlap* if it has a factor of the form $aXaXa$, where a is a letter and X is a possibly empty word. A classical theorem of Thue says that the fixed point of [equation \(6\)](#) is overlap-free. In particular it contains no cube XXX with $X \neq \emptyset$. This combinatorial fact is not visible in a single arithmetic formula, but it is central to the sequence's original role in combinatorics on words. It also illustrates why a complete atlas needs both formulae and substitution structure.

3 Digit, floor, fractional-part, and trigonometric formulae

3.1 Extracting a binary digit

For every $j \geq 0$,

$$b_j(n) = \left\lfloor \frac{n}{2^j} \right\rfloor - 2 \left\lfloor \frac{n}{2^{j+1}} \right\rfloor = \left\lfloor \frac{n}{2^j} \right\rfloor \bmod 2. \quad (14)$$

Summing over j yields several exact formulae.

Theorem 3.1 (Floor and fractional-part formulae). *For every $n \geq 0$,*

$$s_2(n) = n - \sum_{j \geq 1} \left\lfloor \frac{n}{2^j} \right\rfloor, \quad (15)$$

$$= \sum_{j \geq 1} \frac{[n]_{2^j}}{2^j}, \quad (16)$$

$$= \sum_{j \geq 1} \left\{ \frac{n}{2^j} \right\}. \quad (17)$$

The floor sum is finite; the latter two series converge geometrically after $2^j > n$.

Proof. Summing [equation \(14\)](#) over $j \geq 0$ telescopes:

$$\sum_{j \geq 0} b_j(n) = \sum_{j \geq 0} \left\lfloor \frac{n}{2^j} \right\rfloor - 2 \sum_{j \geq 0} \left\lfloor \frac{n}{2^{j+1}} \right\rfloor = n - \sum_{j \geq 1} \left\lfloor \frac{n}{2^j} \right\rfloor.$$

Since $n = 2^j \lfloor n/2^j \rfloor + [n]_{2^j}$ and $\sum_{j \geq 1} n/2^j = n$, substitution gives the remainder and fractional-part forms. $\square$

Consequently

$$\varepsilon_n = (-1)^{n - \sum_{j \geq 1} \lfloor n/2^j \rfloor}. \quad (18)$$

3.2 Products of local digit signs

Because $1 - 2b_j(n) = (-1)^{b_j(n)}$,

$$\varepsilon_n = \prod_{j \geq 0} (1 - 2b_j(n)). \quad (19)$$

This is finite because all sufficiently high digits vanish. Replacing a bit by a cosine gives

$$\varepsilon_n = \prod_{j \geq 0} \cos\left(\pi \left\lfloor \frac{n}{2^j} \right\rfloor\right), \quad (20)$$

again with all but finitely many factors equal to 1. A single trigonometric formula is

$$\varepsilon_n = \cos(\pi s_2(n)), \quad \tau(n) = \sin^2\left(\frac{\pi}{2} s_2(n)\right). \quad (21)$$

3.3 A sinc sign interpolation

Define $\text{sinc } x = \sin x/x$ for $x \neq 0$ and $\text{sinc } 0 = 1$, and put

$$\Phi(x) = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi x}{2^j}\right). \quad (22)$$

The product converges locally uniformly because $1 - \text{sinc } y = \mathcal{O}(y^2)$ near 0.

Proposition 3.2 (Signs between consecutive integers). *For every integer $N \geq 0$ and every $x \in (N, N+1)$,*

$$\text{sgn } \Phi(x) = \varepsilon_N. \quad (23)$$

In particular,

$$\varepsilon_N = \text{sgn } \Phi\left(N + \frac{1}{2}\right). \quad (24)$$

Proof. For positive x the denominator of every sinc factor is positive. Thus its sign is the sign of $\sin(\pi x/2^j)$. For fixed $x \in (N, N+1)$, the number of negative factors equals, modulo 2, the sum of the binary digits of N . One way to see this is to compare signs recursively: from

$$\Phi(2x) = \text{sinc}(2\pi x)\Phi(x)$$

and the sign of $\sin(2\pi x)$ on half-unit intervals, the two interval signs above N are ε_N and $-\varepsilon_N$, exactly as in [equation \(5\)](#). Since the initial interval $(0, 1)$ has positive sign, induction proves the claim. $\square$

Remark 3.3. The zeros of Φ at the positive integers make it a continuous analytic carrier of a discontinuous symbolic pattern. The formula does not interpolate ε_N by value; it interpolates it by the sign on complementary intervals.

4 Factorial valuations, central binomials, Catalan numbers, and carries

4.1 Legendre's identity in base two

Theorem 4.1 (Legendre). *For every $n \geq 0$,*

$$v_2(n!) = \sum_{j \geq 1} \left\lfloor \frac{n}{2^j} \right\rfloor = n - s_2(n). \quad (25)$$

Therefore

$$\varepsilon_n = (-1)^{n-v_2(n!)} = (-1)^{n+v_2(n!)}. \quad (26)$$

Proof. Every multiple of 2 contributes at least one factor 2, every multiple of 4 contributes one additional factor, and so on. This gives the floor sum. The second equality is [equation \(15\)](#). $\square$

4.2 A single central binomial coefficient

Theorem 4.2 (Central-binomial valuation). *For every $n \geq 0$,*

$$v_2 \binom{2n}{n} = s_2(n). \quad (27)$$

Consequently

$$\boxed{\varepsilon_n = (-1)^{v_2 \binom{2n}{n}}, \quad \tau(n) = \left[v_2 \binom{2n}{n} \right]_2.} \quad (28)$$

Proof. By [equation \(25\)](#),

$$v_2 \binom{2n}{n} = v_2((2n)!) - 2v_2(n!) = (2n - s_2(2n)) - 2(n - s_2(n)).$$

Since $s_2(2n) = s_2(n)$, this is $s_2(n)$. $\square$

This formula is particularly economical: a statistic of all binary digits is recovered from the 2-adic order of one classical integer.

4.3 Catalan numbers

Let

$$\text{Cat}_n = \frac{1}{n+1} \binom{2n}{n}.$$

Proposition 4.3 (Catalan valuation). *For $n \geq 0$,*

$$v_2(\text{Cat}_n) = s_2(n+1) - 1. \quad (29)$$

Hence for $n \geq 0$,

$$\varepsilon_{n+1} = -(-1)^{v_2(\text{Cat}_n)}, \quad \tau(n+1) = 1 - [v_2(\text{Cat}_n)]_2. \quad (30)$$

Proof. By [theorem 4.2](#), $v_2(\text{Cat}_n) = s_2(n) - v_2(n+1)$. If $r = v_2(n+1)$, then adding 1 to n replaces exactly r trailing 1's by 0's and then changes one 0 to 1. Thus $s_2(n+1) = s_2(n) - r + 1$. $\square$

4.4 Kummer's theorem and the carry-corrected addition law

Theorem 4.4 (Kummer in base two). *The number of carries produced when adding a and b in base two is*

$$c_2(a, b) = v_2 \binom{a+b}{a} = s_2(a) + s_2(b) - s_2(a+b). \quad (31)$$

Proof. The valuation identity follows from [equation \(25\)](#):

$$\begin{aligned} v_2 \binom{a+b}{a} &= a + b - s_2(a+b) - (a - s_2(a)) - (b - s_2(b)) \\ &= s_2(a) + s_2(b) - s_2(a+b). \end{aligned}$$

Each binary carry reduces the total digit sum by exactly one relative to the no-carry sum, counted with propagation; this gives the carry interpretation. $\square$

Theorem 4.5 (Carry-corrected multiplicativity). *For all $a, b \geq 0$,*

$$\boxed{\varepsilon_{a+b} = \varepsilon_a \varepsilon_b (-1)^{v_2 \binom{a+b}{a}}}. \quad (32)$$

Equivalently,

$$\tau(a+b) \equiv \tau(a) + \tau(b) + v_2 \binom{a+b}{a} \pmod{2}. \quad (33)$$

Proof. Reduce [equation \(31\)](#) modulo 2 and exponentiate -1 . $\square$

Example 4.6. Take $a = 13 = (1101)_2$ and $b = 7 = (0111)_2$. Then $s_2(a) = 3$, $s_2(b) = 3$, $a + b = 20 = (10100)_2$, and $s_2(a + b) = 2$. Hence $v_2 \binom{20}{13} = 4$ and $\varepsilon_{20} = \varepsilon_{13} \varepsilon_7 (-1)^4 = (-1)(-1) = +1$, as expected.

4.5 Subtraction and borrow counts

For $0 \leq b \leq a$, Kummer also gives

$$v_2 \binom{a}{b} = s_2(b) + s_2(a - b) - s_2(a), \quad (34)$$

which counts borrows in the subtraction $a - b$. Thus

$$\varepsilon_{a-b} = \varepsilon_a \varepsilon_b (-1)^{v_2 \binom{a}{b}}. \quad (35)$$

This is the subtraction analogue of [equation \(32\)](#).

5 Pascal’s triangle, Lucas support, and log-free row formulae

5.1 Odd entries and the Boolean submask lattice

For nonnegative integers k, n , write $k \preceq_2 n$ when every 1-bit of k occurs in a position where n also has a 1-bit. Equivalently, $k \& \neg n = 0$.

Theorem 5.1 (Lucas in base two). *For $0 \leq k \leq n$,*

$$\binom{n}{k} \equiv 1 \pmod{2} \iff k \preceq_2 n. \quad (36)$$

Consequently the number of odd entries in row n of Pascal’s triangle is

$$\mathcal{O}(n) := \# \left\{ 0 \leq k \leq n : \binom{n}{k} \text{ is odd} \right\} = 2^{s_2(n)}. \quad (37)$$

Proof. Lucas’s congruence writes

$$\binom{n}{k} \equiv \prod_{j \geq 0} \binom{b_j(n)}{b_j(k)} \pmod{2}.$$

A factor vanishes precisely when $b_j(k) = 1$ and $b_j(n) = 0$. Therefore odd entries are indexed by subsets of the $s_2(n)$ occupied bit positions, giving $2^{s_2(n)}$ choices. $\square$

The support set

$$\mathcal{B}(n) = \{k : k \preceq_2 n\} \quad (38)$$

is a Boolean lattice of rank $s_2(n)$. Its Möbius function from 0 to k is

$$\mu_{\mathcal{B}(n)}(0, k) = (-1)^{s_2(k)} = \varepsilon_k. \quad (39)$$

Thus the signed Thue–Morse value is literally the Möbius sign of the binary submask lattice.

5.2 The binomial–logarithm formula and its exact variants

Taking a base-two logarithm in [equation \(37\)](#) gives

$$s_2(n) = \log_2 \mathcal{O}(n), \quad \tau(n) = [\log_2 \mathcal{O}(n)]_2, \quad \varepsilon_n = (-1)^{\log_2 \mathcal{O}(n)}. \quad (40)$$

This is exact because $\mathcal{O}(n)$ is a power of two. The odd-count itself can be written without a parity predicate:

$$\mathcal{O}(n) = \sum_{k=0}^n \frac{1 - (-1)^{\binom{n}{k}}}{2}, \quad (41)$$

$$= \sum_{k=0}^n \sin^2\left(\frac{\pi}{2} \binom{n}{k}\right), \quad (42)$$

$$= \frac{1}{2} \left(n + 1 - \sum_{k=0}^n \cos\left(\pi \binom{n}{k}\right) \right). \quad (43)$$

These forms explain the binomial–logarithm identity in the companion Fabius-function exposition.

5.3 A log-free formula modulo three

Let $[a]_q \in \{0, 1, \dots, q-1\}$ denote the least nonnegative residue of the integer a modulo q . Since $2^r \equiv 1$ or $2 \pmod{3}$ according as r is even or odd, [equation \(37\)](#) immediately removes the logarithm.

Theorem 5.2 (Log-free Pascal formula). *For every $n \geq 0$,*

$$\tau(n) = [\mathcal{O}(n)]_3 - 1, \quad \varepsilon_n = 3 - 2[\mathcal{O}(n)]_3. \quad (44)$$

Here $[\mathcal{O}(n)]_3$ is always 1 or 2, never 0.

Proof. By [equation \(37\)](#), $[\mathcal{O}(n)]_3 = [2^{s_2(n)}]_3$. This residue equals 1 when $s_2(n)$ is even and 2 when it is odd. $\square$

There is also a formula involving a single signed row statistic. Define

$$\Sigma(n) = \sum_{k=1}^n (-1)^{\binom{n}{k}}, \quad (45)$$

with the empty sum $\Sigma(0) = 0$.

Proposition 5.3 (Signed-row residue). *For every $n \geq 0$,*

$$\tau(n) = [2n + \Sigma(n)]_3. \quad (46)$$

Equivalently,

$$\tau(n) = \left[2n + \sum_{k=1}^n \cos\left(\pi \binom{n}{k}\right) \right]_3. \quad (47)$$

Proof. Among $k = 1, \dots, n$, exactly $\mathcal{O}(n) - 1$ binomial coefficients are odd. Therefore

$$\Sigma(n) = (n - \mathcal{O}(n) + 1) - (\mathcal{O}(n) - 1) = n + 2 - 2\mathcal{O}(n).$$

Hence $2n + \Sigma(n) \equiv 2 - 2\mathcal{O}(n) \pmod{3}$, which is 0 when $\mathcal{O}(n) \equiv 1$ and 1 when $\mathcal{O}(n) \equiv 2$. $\square$

5.4 A one-parameter row statistic

For an indeterminate or complex parameter u , define

$$R_u(n) = \sum_{k=0}^n u^{\binom{n}{k} \bmod 2}. \quad (48)$$

Identity 5.4 (Pascal row partition function).

$$R_u(n) = n + 1 + (u - 1)2^{s_2(n)}. \quad (49)$$

Proof. Every even entry contributes 1, and every odd entry contributes u . There are $2^{s_2(n)}$ odd entries. $\square$

Specializations organize several earlier formulae:

$$R_0(n) = n + 1 - 2^{s_2(n)} \quad \text{counts even entries,}$$

$$R_1(n) = n + 1,$$

$$R_{-1}(n) = n + 1 - 2^{s_2(n)+1} = \sum_{k=0}^n (-1)^{\binom{n}{k}},$$

$$\left. \frac{\partial}{\partial u} R_u(n) \right|_{u=1} = 2^{s_2(n)} \quad \text{counts odd entries.}$$

The parameter u is therefore a compact generating device for all statistics that distinguish only the parity of each Pascal entry.

5.5 Nested binomial parity probes

Lucas’s theorem supplies a universal probe for binary digits of any integer $M \geq 0$:

$$b_r(M) \equiv \binom{M}{2^r} \pmod{2}. \quad (50)$$

Taking $M = \binom{n}{k}$ yields nested binomial coefficients

$$b_r\left(\binom{n}{k}\right) \equiv \binom{\binom{n}{k}}{2^r} \pmod{2}. \quad (51)$$

Thus every binary digit of every Pascal entry can be represented by the parity of another binomial coefficient. Summing over r gives an exact, though not usually economical, formula

$$s_2\left(\binom{n}{k}\right) = \sum_{r \geq 0} \left[\binom{\binom{n}{k}}{2^r} \right]_2. \quad (52)$$

This hierarchy is useful when a formal library already has Lucas congruences but not bit-extraction lemmas for arbitrary integers.

5.6 Power-of-two rows and cyclotomic sparsity

For $m \geq 0$, all interior coefficients of $(1 + x)^{2^m}$ are even, so over $\mathbb{F}_2$

$$(1 + x)^{2^m} = 1 + x^{2^m}. \quad (53)$$

More generally, if $n = \sum_{j \in J(n)} 2^j$, then

$$(1 + x)^n \equiv \prod_{j \in J(n)} (1 + x^{2^j}) \pmod{2}. \quad (54)$$

The coefficient support on the right is exactly the submask set $\mathcal{B}(n)$. This polynomial form of Lucas’s theorem will reappear in the sparse Prouhet identities of [section 12](#).

6 Multinomial parity and higher-dimensional support

6.1 Odd multinomial coefficients

For $r \geq 2$ and $n_1 + \dots + n_r = n$, write

$$\binom{n}{n_1, \dots, n_r} = \frac{n!}{n_1! \dots n_r!}.$$

Theorem 6.1 (Odd multinomial support). *The number of odd multinomial coefficients in the full r -nomial layer of weight n is*

$$\mathcal{O}_r(n) := \# \left\{ (n_1, \dots, n_r) : \sum n_i = n, \binom{n}{n_1, \dots, n_r} \text{ odd} \right\} = r^{s_2(n)}. \quad (55)$$

More precisely, the parity-support polynomial is

$$\sum_{\substack{n_1 + \dots + n_r = n \\ \binom{n}{n_1, \dots, n_r} \text{ odd}}} x_1^{n_1} \dots x_r^{n_r} = \prod_{j \in J(n)} (x_1^{2^j} + \dots + x_r^{2^j}). \quad (56)$$

Proof. In characteristic two,

$$(x_1 + \dots + x_r)^n = \prod_{j \in J(n)} (x_1 + \dots + x_r)^{2^j} = \prod_{j \in J(n)} (x_1^{2^j} + \dots + x_r^{2^j}).$$

Every occupied binary position of n must be assigned independently to one of the r components. There are $r^{s_2(n)}$ assignments, and no carries are permitted among the component digit sums. $\square$

Corollary 6.2 (Multinomial recovery of Thue–Morse). *For any fixed integer $r \geq 2$,*

$$\tau(n) = [\log_r \mathcal{O}_r(n)]_2, \quad \varepsilon_n = (-1)^{\log_r \mathcal{O}_r(n)}. \quad (57)$$

When $r \equiv -1 \pmod{3}$, the logarithm can again be removed:

$$\tau(n) = [\mathcal{O}_r(n)]_3 - 1, \quad \varepsilon_n = 3 - 2[\mathcal{O}_r(n)]_3. \quad (58)$$

Proof. The logarithmic formula is immediate from [equation \(55\)](#). If $r \equiv 2 \pmod{3}$, then $r^{s_2(n)}$ alternates between residues 1 and 2 according to the parity of $s_2(n)$. $\square$

6.2 Valuation and carry interpretation

Legendre’s identity gives

$$v_2 \binom{n}{n_1, \dots, n_r} = \sum_{i=1}^r s_2(n_i) - s_2(n). \quad (59)$$

This is zero precisely when adding the n_i in base two produces no carry. Thus [theorem 6.1](#) is a carry-free allocation theorem: each occupied bit of n chooses exactly one component.

6.3 A multivariate Boolean lattice

The support in [equation \(56\)](#) is the Cartesian product of $s_2(n)$ copies of an r -element choice set. For $r = 2$ it is the Boolean submask lattice. For general r it is the discrete cube $\{1, \dots, r\}^{s_2(n)}$. This makes the exponent $r^{s_2(n)}$ conceptually inevitable and suggests higher-dimensional Walsh transforms adapted to multinomial support.

7 Evil and odious numbers, counting, and exact summatory formulae

An integer is called *evil* when $\tau(n) = 0$ and *odious* when $\tau(n) = 1$. Let E_j and O_j be the j th evil and odious nonnegative integers, indexed from $j = 0$.

Theorem 7.1 (Exact enumerators). *For every $j \geq 0$,*

$$\boxed{E_j = 2j + \tau(j), \quad O_j = 2j + 1 - \tau(j).} \quad (60)$$

In particular,

$$O_j - E_j = 1 - 2\tau(j) = \varepsilon_j. \quad (61)$$

Proof. The pair $(2j, 2j + 1)$ has signs $(\varepsilon_j, -\varepsilon_j)$ by [equation \(5\)](#). Therefore exactly one member of the pair is evil and one is odious. If $\tau(j) = 0$, they occur in the order $(E_j, O_j) = (2j, 2j + 1)$; if $\tau(j) = 1$, the order is reversed. $\square$

7.1 Exact prefix sums

Define the signed prefix sum

$$S(N) = \sum_{n=0}^{N-1} \varepsilon_n, \quad S(0) = 0. \quad (62)$$

Theorem 7.2 (Exact prefix discrepancy). *For every $q \geq 0$,*

$$S(2q) = 0, \quad S(2q + 1) = \varepsilon_q. \quad (63)$$

Hence $S(N) \in \{-1, 0, 1\}$ for every N .

Proof. Pair consecutive terms: $\varepsilon_{2j} + \varepsilon_{2j+1} = \varepsilon_j - \varepsilon_j = 0$. An odd prefix has the additional term $\varepsilon_{2q} = \varepsilon_q$. $\square$

Let

$$E(N) = \#\{0 \leq n < N : \tau(n) = 0\}, \quad O(N) = \#\{0 \leq n < N : \tau(n) = 1\}.$$

Then

$$E(N) = \frac{N + S(N)}{2}, \quad O(N) = \frac{N - S(N)}{2}. \quad (64)$$

Thus every prefix has exactly balanced evil and odious counts when N is even, and imbalance one when N is odd.

7.2 Interval discrepancy

For integers $A, B \geq 0$, put

$$D(A, B) = S(B) - S(A).$$

When $A < B$, this is the signed interval sum $D(A, B) = \sum_{n=A}^{B-1} \varepsilon_n$.

Corollary 7.3 (Uniform interval bound). *For all integers $0 \leq A < B$,*

$$|D(A, B)| \leq 2. \quad (65)$$

More precisely, for $a, b \geq 0$,

$$D(2a, 2b) = 0, \quad (66)$$

$$D(2a, 2b + 1) = \varepsilon_b, \quad (67)$$

$$D(2a + 1, 2b) = -\varepsilon_a, \quad (68)$$

$$D(2a + 1, 2b + 1) = \varepsilon_b - \varepsilon_a. \quad (69)$$

Thus opposite-parity endpoints give $|D(A, B)| = 1$; even endpoints give zero; and odd endpoints give one of $-2, 0, 2$.

Proof. Substitute the exact prefix values $S(2q) = 0$ and $S(2q + 1) = \varepsilon_q$ into $D(A, B) = S(B) - S(A)$. $\square$

Remark 7.4. This exact discrepancy bound is special to unweighted sums. Exponential sums $\sum_{n < N} \varepsilon_n e^{2\pi i n \alpha}$ have much subtler growth; the sharp uniform exponent is discussed in [section 15](#).

8 Finite generating products and their algebra

8.1 The coefficient product

Theorem 8.1 (Finite product identity). *For every $m \geq 0$,*

$$P_m(z) := \sum_{n=0}^{2^m-1} \varepsilon_n z^n = \prod_{j=0}^{m-1} (1 - z^{2^j}). \quad (70)$$

Proof. Expanding the product means choosing either 1 or $-z^{2^j}$ from each factor. A choice set $J \subseteq \{0, \dots, m-1\}$ contributes $(-1)^{|J|} z^{\sum_{j \in J} 2^j}$. Every exponent $0 \leq n < 2^m$ has a unique binary choice set, and the sign is $(-1)^{s_2(n)}$. $\square$

The product obeys two complementary recursions:

$$P_{m+1}(z) = (1 - z^{2^m}) P_m(z), \quad (71)$$

$$P_m(z) = (1 - z) P_{m-1}(z^2) \quad (m \geq 1). \quad (72)$$

The first appends the highest bit; the second separates the least significant bit.

8.2 Reciprocity and cyclotomic factorization

Let $D_m = 2^m - 1 = \deg P_m$. Then

$$z^{D_m} P_m(z^{-1}) = (-1)^m P_m(z). \quad (73)$$

Equating coefficients gives the reflection rule

$$\varepsilon_{D_m-n} = (-1)^m \varepsilon_n, \quad 0 \leq n \leq D_m. \quad (74)$$

This is the complement-all-bits symmetry of an m -bit word.

Factoring each $1 - z^{2^j}$ into dyadic cyclotomic factors gives

$$P_m(z) = (1 - z)^m \prod_{r=0}^{m-2} (1 + z^{2^r})^{m-1-r}. \quad (75)$$

Therefore the complete zero set consists of roots of unity of order a power of two. The multiplicity of a primitive 2^{r+1} st root is $m - 1 - r$, while $z = 1$ has multiplicity m .

8.3 Coefficient extraction and binary uniqueness

The finite product yields several exact coefficient formulae:

$$\varepsilon_n = [z^n] \prod_{j \geq 0} (1 - z^{2^j}), \quad (76)$$

$$\varepsilon_n = \sum_{J \subseteq \{0, \dots, m-1\}} (-1)^{|J|} \mathbf{1} \left(n = \sum_{j \in J} 2^j \right) \quad (0 \leq n < 2^m), \quad (77)$$

$$\tau(n) = \frac{1}{2} (1 - [z^n] P_m(z)) \quad (0 \leq n < 2^m). \quad (78)$$

The sum in [equation \(77\)](#) has exactly one nonzero term. Its importance is structural rather than computational: it realizes Thue–Morse as the signed incidence function of binary subset sums.

8.4 Weighted digit sums

A useful umbrella product is

$$F_u(z) = \sum_{n \geq 0} u^{s_2(n)} z^n = \prod_{j \geq 0} (1 + u z^{2^j}), \quad |z| < 1. \quad (79)$$

It satisfies

$$F_u(z) = (1 + uz) F_u(z^2). \quad (80)$$

The cases $u = 1$ and $u = -1$ are

$$F_1(z) = \frac{1}{1 - z}, \quad F_{-1}(z) = E(z). \quad (81)$$

Differentiating with respect to u at $u = 1$ gives the ordinary generating function for the binary digit sum:

$$\sum_{n \geq 0} s_2(n) z^n = \frac{1}{1 - z} \sum_{j \geq 0} \frac{z^{2^j}}{1 + z^{2^j}}. \quad (82)$$

Higher u -derivatives generate the falling-factorial moments $s_2(n)^r$.

9 Infinite products, Mahler equations, and coefficient recurrences

9.1 The signed analytic generating function

For $|z| < 1$, the finite products converge locally uniformly to

$$E(z) = \sum_{n \geq 0} \varepsilon_n z^n = \prod_{j=0}^{\infty} (1 - z^{2^j}). \quad (83)$$

The first factor separates:

$$E(z) = (1 - z) E(z^2), \quad E(0) = 1. \quad (84)$$

Iterating r times gives the lacunary quotient

$$\frac{E(z)}{E(z^{2^r})} = \prod_{j=0}^{r-1} (1 - z^{2^j}) = P_r(z). \quad (85)$$

As $r \rightarrow \infty$, $E(z^{2^r}) \rightarrow 1$, recovering the infinite product.

9.2 The zero–one generating function

Using [equation \(2\)](#),

$$T(z) = \sum_{n \geq 0} \tau(n) z^n = \frac{1}{2(1-z)} - \frac{1}{2} E(z). \quad (86)$$

Separating even and odd indices gives the inhomogeneous Mahler equation

$$T(z) = (1-z)T(z^2) + \frac{z}{1-z^2}. \quad (87)$$

Indeed,

$$\begin{aligned} T(z) &= \sum_{n \geq 0} \tau(2n) z^{2n} + \sum_{n \geq 0} \tau(2n+1) z^{2n+1} \\ &= T(z^2) + z \left(\frac{1}{1-z^2} - T(z^2) \right). \end{aligned}$$

9.3 Logarithmic derivative and the ruler function

Taking a logarithm in [equation \(83\)](#),

$$\begin{aligned} \log E(z) &= \sum_{j \geq 0} \log(1 - z^{2^j}) \\ &= - \sum_{j \geq 0} \sum_{q \geq 1} \frac{z^{q2^j}}{q} = - \sum_{r \geq 1} \frac{a_r}{r} z^r, \end{aligned} \quad (88)$$

where

$$a_r = \sum_{2^j | r} 2^j = 2^{v_2(r)+1} - 1. \quad (89)$$

Thus

$$\frac{zE'(z)}{E(z)} = - \sum_{r \geq 1} a_r z^r. \quad (90)$$

The coefficient a_r is an affine function of the ruler sequence $2^{v_2(r)}$.

Theorem 9.1 (Convolution recurrence). *With $\varepsilon_0 = 1$, the signed Thue–Morse coefficients satisfy*

$$n\varepsilon_n = - \sum_{r=1}^n (2^{v_2(r)+1} - 1) \varepsilon_{n-r} \quad (n \geq 1). \quad (91)$$

Proof. Multiply [equation \(90\)](#) by $E(z)$ and compare the coefficient of z^n . □

This recurrence is arithmetically very different from the dyadic recursion: it computes ε_n from all preceding values, weighted by a valuation sequence.

9.4 Bell-polynomial formula

Let $\mathcal{B}_n(x_1, \dots, x_n)$ be the complete exponential Bell polynomial, defined by

$$\exp\left(\sum_{r \geq 1} x_r \frac{z^r}{r!}\right) = \sum_{n \geq 0} \mathcal{B}_n(x_1, \dots, x_n) \frac{z^n}{n!}. \quad (92)$$

Set

$$x_r = -(r-1)!(2^{v_2(r)+1} - 1). \quad (93)$$

Then [equation \(88\)](#) gives the exact formula

$$\varepsilon_n = \frac{1}{n!} \mathcal{B}_n(x_1, \dots, x_n). \quad (94)$$

This is useful when a symbolic algebra system has native Bell-polynomial support.

9.5 A determinant formula

For $n \geq 1$, define the lower-Hessenberg matrix

$$H_n = \begin{pmatrix} a_1 & 1 & 0 & 0 & \cdots & 0 \\ a_2 & a_1 & 2 & 0 & \cdots & 0 \\ a_3 & a_2 & a_1 & 3 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & \ddots & 0 \\ a_{n-1} & a_{n-2} & \cdots & a_2 & a_1 & n-1 \\ a_n & a_{n-1} & \cdots & a_3 & a_2 & a_1 \end{pmatrix}, \quad a_r = 2^{v_2(r)+1} - 1. \quad (95)$$

Theorem 9.2 (Hessenberg determinant). *For every $n \geq 1$,*

$$\varepsilon_n = \frac{(-1)^n}{n!} \det H_n. \quad (96)$$

In particular,

$$|\det H_n| = n!. \quad (97)$$

Proof. Expansion of a lower-Hessenberg determinant along the last row gives exactly the coefficient recurrence associated with $E(z) = \exp(-\sum_{r \geq 1} a_r z^r / r)$. Equivalently, this is the standard determinantal form of complete Bell polynomials applied to [equation \(94\)](#). The absolute-value statement follows from $\varepsilon_n \in \{\pm 1\}$. $\square$

9.6 Euler-transform form

The product [equation \(83\)](#) is also an Euler transform with exponents supported at powers of two:

$$E(z) = \prod_{r \geq 1} (1 - z^r)^{c_r}, \quad c_r = \begin{cases} 1, & r \text{ is a power of 2,} \\ 0, & \text{otherwise.} \end{cases} \quad (98)$$

The logarithmic derivative of a general Euler product is a divisor sum. Here that divisor sum collapses to [equation \(89\)](#). This explains why the ruler function is the natural convolution partner of Thue–Morse.

10 Algebraic avatars in characteristic two and over the integers

10.1 The Artin–Schreier equation

Reduce the coefficients of $T(z)$ modulo 2 and regard

$$\Theta(z) = \sum_{n \geq 0} \tau(n)z^n \in \mathbb{F}_2[[z]].$$

Since $\Theta(z^2) = \Theta(z)^2$ in characteristic two, reducing [equation \(87\)](#) gives

$$\Theta(z) = (1+z)\Theta(z)^2 + \frac{z}{1+z^2}. \quad (99)$$

After clearing denominators,

$$(1+z)^3\Theta(z)^2 + (1+z)^2\Theta(z) + z = 0. \quad (100)$$

If

$$Y(z) = (1+z)\Theta(z), \quad (101)$$

then [equation \(100\)](#) becomes the Artin–Schreier equation

$$Y(z)^2 + Y(z) = \frac{z}{1+z}. \quad (102)$$

Thus the ordinary generating series of the zero–one sequence is algebraic of degree two over $\mathbb{F}_2(z)$.

Remark 10.1. Christol’s theorem says that, over a finite field of characteristic p , a coefficient sequence is p -automatic if and only if its generating series is algebraic over the corresponding rational function field. Here $p = 2$, and both directions are visible directly: the two-state automaton gives the recursion, while the recursion gives the explicit quadratic equation.

10.2 An integer algebraic lift

Define integers A_n by $A_0 = 0$, $A_1 = 1$, and for $n \geq 2$,

$$nA_n = (5 - 2n)A_{n-1} + 3(n-1)A_{n-2} + 1. \quad (103)$$

Their generating function is

$$A(z) = \sum_{n \geq 0} A_n z^n = \frac{1}{2(1-z)} \left(\sqrt{\frac{1+3z}{1-z}} - 1 \right). \quad (104)$$

It satisfies

$$(1-z)^3 A(z)^2 + (1-z)^2 A(z) = z. \quad (105)$$

Proposition 10.2 (Parity lift). *For every $n \geq 0$,*

$$A_n \equiv \tau(n) \pmod{2}. \quad (106)$$

Proof. Reducing [equation \(105\)](#) modulo 2 changes $1 - z$ to $1 + z$ and gives exactly [equation \(100\)](#). Both formal solutions have constant term zero. The coefficient of the first unknown term appears linearly with unit coefficient, so the solution in $\mathbb{F}_2[[z]]$ is unique. Therefore $A(z) \equiv \Theta(z) \pmod{2}$ coefficientwise. $\square$

Remark 10.3. The lift is not a new definition of Thue–Morse; it is an integer sequence whose parity shadow is Thue–Morse. Its value lies in transferring questions about automatic parity to an algebraic generating function over characteristic zero. The square-root singularity also makes asymptotic tools available for A_n .

11 Prouhet cancellation, finite differences, and exact cubature

11.1 The Prouhet partition

For $m \geq 1$, partition $\{0, 1, \dots, 2^m - 1\}$ into

$$\mathcal{E}_m = \{n < 2^m : \varepsilon_n = 1\}, \quad \mathcal{O}_m = \{n < 2^m : \varepsilon_n = -1\}. \quad (107)$$

The finite product has a zero of order m at $z = 1$:

$$P_m(z) = (1 - z)^m \prod_{r=0}^{m-2} (1 + z^{2^r})^{m-1-r}. \quad (108)$$

This is the algebraic source of Prouhet’s equal-sums phenomenon.

Theorem 11.1 (Prouhet cancellation). *For every integer r with $0 \leq r < m$,*

$$\sum_{n=0}^{2^m-1} \varepsilon_n n^r = 0. \quad (109)$$

Equivalently,

$$\sum_{n \in \mathcal{E}_m} n^r = \sum_{n \in \mathcal{O}_m} n^r \quad (0 \leq r < m). \quad (110)$$

The first nonzero moment is

$$\sum_{n=0}^{2^m-1} \varepsilon_n n^m = (-1)^m m! 2^{\binom{m}{2}}. \quad (111)$$

Proof. Apply the operator $(z \frac{d}{dz})^r$ to $P_m(z) = \sum \varepsilon_n z^n$ and evaluate at $z = 1$. The zero of order m forces vanishing for $r < m$. For $r = m$, only the leading term of $P_m(z)$ at $z = 1$ matters:

$$P_m(z) = (-1)^m 2^{0+1+\dots+(m-1)} (z-1)^m + \mathcal{O}((z-1)^{m+1}).$$

The operator $(z d/dz)^m$ has leading part d^m/dz^m at $z = 1$, giving [equation \(111\)](#). $\square$

11.2 The finite-difference identity

For $h \in \mathbb{R}$, let

$$\Delta_h f(x) = f(x+h) - f(x). \quad (112)$$

Theorem 11.2 (Thue–Morse as an iterated difference). *For every function f on which the expressions are defined,*

$$\sum_{n=0}^{2^m-1} \varepsilon_n f(x+n) = (-1)^m \Delta_1 \Delta_2 \cdots \Delta_{2^{m-1}} f(x). \quad (113)$$

Proof. Let $T_h f(x) = f(x+h)$. Then

$$\sum_{n < 2^m} \varepsilon_n T_n = \prod_{j=0}^{m-1} (I - T_{2^j}) = (-1)^m \prod_{j=0}^{m-1} \Delta_{2^j}.$$

Apply both sides to f at x . $\square$

Prouhet cancellation follows immediately because each finite difference lowers the degree of a polynomial by one.

11.3 Exact integral cubature

If f is m times continuously differentiable, then

$$\Delta_h f(x) = h \int_0^1 f'(x + hu) \, du. \quad (114)$$

Iterating gives an exact integral representation.

Theorem 11.3 (Cubature identity). *For $f \in C^m$,*

$$\sum_{n=0}^{2^m-1} \varepsilon_n f(x+n) = (-1)^m 2^{\binom{m}{2}} \int_{[0,1]^m} f^{(m)} \left(x + \sum_{j=0}^{m-1} 2^j u_j \right) \, d\mathbf{u}. \quad (115)$$

Proof. Apply [equation \(114\)](#) successively to [equation \(113\)](#); the product of the step sizes is $\prod_{j=0}^{m-1} 2^j = 2^{\binom{m}{2}}$. $\square$

Corollary 11.4 (Mean-value form). *If $f \in C^m$ on $[x, x + 2^m - 1]$, there exists $\xi \in [x, x + 2^m - 1]$ such that*

$$\sum_{n=0}^{2^m-1} \varepsilon_n f(x+n) = (-1)^m 2^{\binom{m}{2}} f^{(m)}(\xi). \quad (116)$$

The cubature formula is stronger than a vanishing-moment statement: it identifies the complete remainder for arbitrary smooth f .

11.4 All power moments

Define

$$M_{m,r} = \sum_{n=0}^{2^m-1} \varepsilon_n n^r. \quad (117)$$

The exponential generating function in r is

$$\sum_{r \geq 0} M_{m,r} \frac{t^r}{r!} = P_m(e^t) = \prod_{j=0}^{m-1} (1 - e^{2^j t}). \quad (118)$$

Put

$$\varphi(x) = \frac{e^x - 1}{x} = \sum_{q \geq 0} \frac{x^q}{(q+1)!}.$$

Then

$$P_m(e^t) = (-1)^m 2^{\binom{m}{2}} t^m \prod_{j=0}^{m-1} \varphi(2^j t). \quad (119)$$

Theorem 11.5 (Composition formula for every moment). *For $r < m$, $M_{m,r} = 0$. For $r \geq m$,*

$$M_{m,r} = (-1)^m r! 2^{\binom{m}{2}} \sum_{\substack{q_0 + \dots + q_{m-1} = r-m \\ q_j \geq 0}} \prod_{j=0}^{m-1} \frac{2^{jq_j}}{(q_j+1)!}. \quad (120)$$

Proof. Extract the coefficient of t^r from [equation \(119\)](#). $\square$

For example, with $D_m = 2^m - 1$ and $Q_m = 1 + 4 + \dots + 4^{m-1} = (4^m - 1)/3$,

$$M_{m,m} = (-1)^m m! 2^{\binom{m}{2}}, \quad (121)$$

$$M_{m,m+1} = (-1)^m (m+1)! 2^{\binom{m}{2}} \frac{D_m}{2}, \quad (122)$$

$$M_{m,m+2} = (-1)^m (m+2)! 2^{\binom{m}{2}} \left(\frac{D_m^2}{8} + \frac{Q_m}{24} \right). \quad (123)$$

11.5 Centered moments and parity

Let $c_m = (2^m - 1)/2$. By reciprocity,

$$\sum_{n < 2^m} \varepsilon_n e^{(n-c_m)t} = (-2)^m \prod_{j=0}^{m-1} \sinh(2^{j-1}t). \quad (124)$$

The right side has parity m as a function of t .

Corollary 11.6 (Centered parity cancellation). *For every $r \geq 0$,*

$$\sum_{n < 2^m} \varepsilon_n (n - c_m)^r = 0 \quad (125)$$

whenever either $r < m$ or $r \not\equiv m \pmod{2}$.

11.6 Binomial and falling-factorial moments

Set

$$B_{m,r} = \sum_{n < 2^m} \varepsilon_n \binom{n}{r}. \quad (126)$$

Then

$$\sum_{r \geq 0} B_{m,r} u^r = \sum_{n < 2^m} \varepsilon_n (1+u)^n = P_m(1+u). \quad (127)$$

Therefore

$$B_{m,r} = [z^r] \prod_{j=0}^{m-1} (1 - (1+u)^{2^j}). \quad (128)$$

In particular,

$$B_{m,r} = 0 \quad (r < m), \quad B_{m,m} = (-1)^m 2^{\binom{m}{2}}. \quad (129)$$

Since $n^{\underline{r}} = r! \binom{n}{r}$, this also gives all falling-factorial moments. Power moments follow by the Stirling transform

$$n^r = \sum_{q=0}^r \left\{ \begin{matrix} r \\ q \end{matrix} \right\} n^{\underline{q}}. \quad (130)$$

12 Sparse Pascal-support Prouhet identities

The full dyadic block is only one member of a larger family. Fix $N \geq 0$ and let

$$J(N) = \{j : b_j(N) = 1\}, \quad s = |J(N)| = s_2(N), \quad \beta(N) = \sum_{j \in J(N)} j. \quad (131)$$

Theorem 12.1 (Signed submask polynomial).

$$\boxed{\sum_{k \preccurlyeq_2 N} \varepsilon_k z^k = \prod_{j \in J(N)} (1 - z^{2^j}).} \quad (132)$$

Proof. Choose a subset of the occupied bit positions of N . The exponent is the corresponding submask k and the sign is (-1) to the number of chosen bits. $\square$

The right side has a zero of order s at $z = 1$.

Corollary 12.2 (Sparse Prouhet cancellation). For $0 \leq r < s$,

$$\sum_{k \preccurlyeq_2 N} \varepsilon_k k^r = 0, \quad (133)$$

and the first nonzero moment is

$$\sum_{k \preccurlyeq_2 N} \varepsilon_k k^s = (-1)^s s! 2^{\beta(N)}. \quad (134)$$

The exact finite-difference form is

$$\sum_{k \preccurlyeq_2 N} \varepsilon_k f(x + k) = (-1)^s \prod_{j \in J(N)} \Delta_{2^j} f(x). \quad (135)$$

Thus every Pascal row carries a Prouhet system whose degree of cancellation is the number of odd binary positions in its row index. The ordinary dyadic Prouhet partition is the special case $N = 2^m - 1$.

12.1 Unsigned and signed support together

The unsigned support polynomial from Lucas's theorem is

$$Q_N(z) = \sum_{k \preccurlyeq_2 N} z^k = \prod_{j \in J(N)} (1 + z^{2^j}), \quad (136)$$

whereas the signed one is obtained by replacing every plus by a minus. The pair

$$Q_N(z), \quad \tilde{Q}_N(z) = \sum_{k \preccurlyeq_2 N} \varepsilon_k z^k \quad (137)$$

encodes the zeta and Möbius transforms of the same Boolean lattice.

13 Iterated prefix sums and the Fabius bridge

13.1 Discrete antiderivatives

Define inclusive iterated prefix sums by

$$S_n^{(0)} = \varepsilon_n, \quad S_n^{(k+1)} = \sum_{j=0}^n S_j^{(k)}. \quad (138)$$

The first level is explicit:

$$S_{2m}^{(1)} = \varepsilon_m, \quad S_{2m+1}^{(1)} = 0. \quad (139)$$

Theorem 13.1 (Binomial kernel for repeated summation). *For $k \geq 1$,*

$$S_n^{(k)} = \sum_{j=0}^n \binom{n-j+k-1}{k-1} \varepsilon_j. \quad (140)$$

Its ordinary generating function is

$$\sum_{n \geq 0} S_n^{(k)} z^n = \frac{E(z)}{(1-z)^k}. \quad (141)$$

Proof. One inclusive prefix multiplies an ordinary generating function by $(1-z)^{-1}$. Iterate and expand $(1-z)^{-k} = \sum_{r \geq 0} \binom{r+k-1}{k-1} z^r$. $\square$

13.2 Terminal zero runs in dyadic blocks

Theorem 13.2 (Dyadic terminal zero run). *If $1 \leq k \leq r$, then*

$$S_{2^r-k+i}^{(k)} = 0, \quad 0 \leq i < k, \quad (142)$$

while

$$S_{2^r-k-1}^{(k)} = (-1)^{r-k}, \quad S_{2^r}^{(k)} = -1. \quad (143)$$

Proof. Below degree 2^r , replace $E(z)$ in [equation \(141\)](#) by $P_r(z)$. Then

$$\frac{P_r(z)}{(1-z)^k} = (1-z)^{r-k} \prod_{j=0}^{r-1} (1+z+\dots+z^{2^j-1}) \quad (144)$$

is a polynomial of degree $2^r - k - 1$. Its coefficients in degrees $2^r - k, \dots, 2^r - 1$ therefore vanish. The leading coefficient gives the first boundary value, and the first omitted factor $1 - z^{2^r}$ gives the coefficient at degree 2^r . $\square$

13.3 Why the same product appears in Fabius theory

The exact algebraic chain is

$$\text{binary character} \implies \prod_{j < r} (1 - z^{2^j}) \implies \frac{\prod_{j < r} (1 - z^{2^j})}{(1-z)^k} \implies \text{iterated-prefix coefficients}. \quad (145)$$

After suitable normalization by products of the dyadic step sizes, coefficient grids and their polygonal or spline interpolants lead to the Fabius function and the Rvachev up-function. On the Fourier side, the same finite product becomes a finite product of sine factors; after rescaling it tends to the sinc product that is the Fourier transform of the up-function. Thus the discrete prefix construction and the analytic Fabius construction are two limits of the same product identity.

The role of Prouhet cancellation is especially transparent. The zero of order k at $z = 1$ creates k vanishing moments, which in turn creates the endpoint flatness needed for smooth compactly supported limits. This is the precise reason Thue–Morse belongs in the Fabius development rather than merely being a coincidental source of coefficients.

14 Walsh and ordinary Fourier transforms

14.1 Walsh transform: a one-point spectrum

Identify $0 \leq n < 2^m$ with its digit vector $\mathbf{b}(n) \in (\mathbb{F}_2)^m$. For $a \in (\mathbb{F}_2)^m$, define the unnormalized Walsh transform

$$\mathcal{W}_m(a) = \sum_{x \in (\mathbb{F}_2)^m} (-1)^{\langle \mathbf{1}, x \rangle} (-1)^{\langle a, x \rangle}. \quad (146)$$

Character orthogonality gives

$$\mathcal{W}_m(a) = \begin{cases} 2^m, & a = \mathbf{1}, \\ 0, & a \neq \mathbf{1}. \end{cases} \quad (147)$$

Thus the Thue–Morse block is maximally simple for the Boolean-group Fourier transform: it is itself one character.

14.2 Ordinary discrete Fourier transform

Let $N = 2^m$ and

$$\widehat{\varepsilon}_m(k) = \sum_{n=0}^{N-1} \varepsilon_n e^{-2\pi i k n / N}, \quad 0 \leq k < N. \quad (148)$$

Evaluation of the generating polynomial gives

$$\widehat{\varepsilon}_m(k) = P_m(e^{-2\pi i k / N}) = \prod_{j=0}^{m-1} (1 - e^{-2\pi i k 2^j / N}). \quad (149)$$

Proposition 14.1 (Exact DFT support and sine product). *If k is even, then $\widehat{\varepsilon}_m(k) = 0$. If k is odd, then*

$$\widehat{\varepsilon}_m(k) = (2i)^m e^{-\pi i k (1 - 2^{-m})} \prod_{r=1}^m \sin\left(\frac{\pi k}{2^r}\right). \quad (150)$$

In particular,

$$|\widehat{\varepsilon}_m(k)| = 2^m \prod_{r=1}^m \left| \sin\left(\frac{\pi k}{2^r}\right) \right| \quad (k \text{ odd}). \quad (151)$$

Proof. The factor with $j = m - 1$ is $1 - (-1)^k$, proving the support statement. For odd k , use $1 - e^{-i\theta} = 2ie^{-i\theta/2} \sin(\theta/2)$ in every factor and sum the phases:

$$\sum_{j=0}^{m-1} \frac{\pi k 2^j}{2^m} = \pi k (1 - 2^{-m}).$$

□

Parseval's identity takes the form

$$\sum_{k=0}^{N-1} |\widehat{\varepsilon}_m(k)|^2 = N^2. \quad (152)$$

All the energy is therefore carried by the $N/2$ odd frequencies.

14.3 A Rademacher sine-sign formula

The ordinary Fourier factors also recover the individual sign. For every $n \geq 0$,

$$\varepsilon_n = \prod_{j \geq 0} \operatorname{sgn} \left(\sin \frac{\pi(n + 1/2)}{2^j} \right). \quad (153)$$

The product is effectively finite because all sufficiently high sine arguments lie in $(0, \pi)$. This is the direct sign-level counterpart of the magnitude product [equation \(151\)](#).

15 Gelfond sums and exact residue-class rarefaction

15.1 The sharp uniform exponent

Put

$$p_m(x) = P_m(e^{2\pi i x}) = \sum_{n < 2^m} \varepsilon_n e^{2\pi i n x}. \quad (154)$$

Then

$$|p_m(x)| = 2^m \prod_{j=0}^{m-1} |\sin(\pi 2^j x)|. \quad (155)$$

The following is Gelfond’s sharp uniform estimate [\[10, 13\]](#).

Theorem 15.1 (Gelfond). *For every $m \geq 1$,*

$$\sup_{x \in \mathbb{R}} |p_m(x)| \leq \frac{2}{\sqrt{3}} 2^{m \log 3 / \log 4} = \frac{2}{\sqrt{3}} 3^{m/2}. \quad (156)$$

Moreover,

$$\sup_{x \in \mathbb{R}} |p_m(x)| \geq 3^{m/2}. \quad (157)$$

Hence the optimal uniform growth exponent is

$$\theta = \frac{\log 3}{\log 4} = 0.792481 \dots \quad (158)$$

Explanation of the lower bound. Let $\omega = e^{2\pi i/3}$. Since $2^j \not\equiv 0 \pmod{3}$ and $|1 - \omega| = |1 - \omega^2| = \sqrt{3}$,

$$|P_m(\omega)| = \prod_{j=0}^{m-1} |1 - \omega^{2^j}| = 3^{m/2}.$$

The upper bound is Gelfond’s nontrivial estimate; its constant is within the factor $2/\sqrt{3}$ of these cubic-root values. $\square$

Binary decomposition of a general interval into dyadic blocks gives the classical consequence

$$\sup_{x \in \mathbb{R}} \left| \sum_{n=0}^{N-1} \varepsilon_n e^{2\pi i n x} \right| \ll N^{\log 3 / \log 4}. \quad (159)$$

The unweighted case $x = 0$ is far smaller because of exact pair cancellation.

15.2 Root-of-unity filters

For a modulus $q \geq 2$, define the signed residue-class sums

$$A_{m,r}^{(q)} = \sum_{\substack{0 \leq n < 2^m \\ n \equiv r \pmod{q}}} \varepsilon_n, \quad 0 \leq r < q. \quad (160)$$

If $\zeta = e^{2\pi i/q}$, the root-of-unity filter gives the exact formula

$$A_{m,r}^{(q)} = \frac{1}{q} \sum_{\ell=0}^{q-1} \zeta^{-r\ell} \prod_{j=0}^{m-1} (1 - \zeta^{\ell 2^j}). \quad (161)$$

This converts arithmetic progressions into finitely many evaluations of the same master product.

15.3 New closed formula modulo three

For $q = 3$, the product can be evaluated completely because the powers of two alternate modulo three.

Theorem 15.2 (Exact dyadic rarefaction modulo three). *Let*

$$A_{m,r} = \sum_{\substack{0 \leq n < 2^m \\ n \equiv r \pmod{3}}} \varepsilon_n. \quad (162)$$

For $s \geq 1$,

$$(A_{2s,0}, A_{2s,1}, A_{2s,2}) = 3^{s-1}(2, -1, -1). \quad (163)$$

For $s \geq 0$,

$$(A_{2s+1,0}, A_{2s+1,1}, A_{2s+1,2}) = 3^s(1, -1, 0). \quad (164)$$

Proof. Let $\omega = e^{2\pi i/3}$. For $m \geq 1$, $P_m(1) = 0$, so

$$A_{m,r} = \frac{1}{3} (\omega^{-r} P_m(\omega) + \omega^{-2r} P_m(\omega^2)). \quad (165)$$

Since $2^j \equiv 1, 2, 1, 2, \dots \pmod{3}$,

$$\begin{aligned} P_{2s}(\omega) &= (1 - \omega)^s (1 - \omega^2)^s = 3^s, \\ P_{2s+1}(\omega) &= 3^s (1 - \omega), \end{aligned}$$

with conjugate formulae at ω^2 . Substitution in [equation \(165\)](#), using $1 + \omega + \omega^2 = 0$, gives the two vectors. $\square$

Example 15.3. The first block vectors are

$$(1, -1, 0), \quad (2, -1, -1), \quad (3, -3, 0), \quad (6, -3, -3), \quad (9, -9, 0).$$

The growth $3^{m/2}$ is exactly the cubic-root lower bound in [equation \(157\)](#); the rarefied sums and the extremal Fourier values are two faces of the same evaluation.

15.4 Matrix recursion for general odd moduli

For odd q , multiplication by 2 permutes the residue classes. Separating even and odd indices gives

$$A_{m+1,r}^{(q)} = A_{m,2^{-1}r}^{(q)} - A_{m,2^{-1}(r-1)}^{(q)}, \quad (166)$$

where residues are taken modulo q and 2^{-1} is the inverse of 2 modulo q . Hence $\mathbf{A}_{m+1} = M_q \mathbf{A}_m$ for an explicit signed permutation-difference matrix. Diagonalizing M_q by the finite Fourier transform reproduces [equation \(161\)](#). This is a convenient finite-dimensional route for exact rarefaction at any fixed odd modulus.

16 Autocorrelation, Riesz products, and the spectral measure

16.1 Autocorrelation coefficients

For $k \in \mathbb{Z}$, the balanced Thue–Morse autocorrelation is

$$\eta(k) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} \varepsilon_n \varepsilon_{n+k}, \quad (167)$$

where the negligible boundary convention for negative indices does not affect the limit. The limit exists by unique ergodicity of the substitution system. It is even, $\eta(-k) = \eta(k)$, and $\eta(0) = 1$.

Theorem 16.1 (Autocorrelation recursion). *For $n \geq 0$,*

$$\eta(2n) = \eta(n), \quad (168)$$

$$\eta(2n+1) = -\frac{1}{2}(\eta(n) + \eta(n+1)). \quad (169)$$

In particular,

$$\eta(1) = -\frac{1}{3}. \quad (170)$$

Proof. For an even displacement, split the average into even and odd starting positions and use $\varepsilon_{2r}\varepsilon_{2r+2n} = \varepsilon_r\varepsilon_{r+n}$ and $\varepsilon_{2r+1}\varepsilon_{2r+2n+1} = \varepsilon_r\varepsilon_{r+n}$. For an odd displacement, the even starts contribute $-\varepsilon_r\varepsilon_{r+n}$ and the odd starts contribute $-\varepsilon_r\varepsilon_{r+n+1}$. The two classes have asymptotic density $1/2$. Putting $n = 0$ in [equation \(169\)](#) gives $\eta(1) = -(1 + \eta(1))/2$. $\square$

The first values are

$$1, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, 0, \frac{1}{3}, 0, -\frac{1}{3}, \frac{1}{6}, 0, -\frac{1}{6}, \frac{1}{3}, -\frac{1}{6}, 0, \frac{1}{6}, \dots$$

16.2 A new functional equation for the one-sided generating series

Define

$$C(z) = \sum_{n \geq 0} \eta(n) z^n, \quad |z| < 1. \quad (171)$$

Theorem 16.2 (Autocorrelation Mahler equation). *The autocorrelation generating series satisfies*

$$2zC(z) = 1 - (1 - z)^2 C(z^2). \quad (172)$$

The identity is valid as a formal power-series identity after the factor z is cleared.

Proof. Split into even and odd coefficients and use [equations \(168\) and \(169\)](#):

$$\begin{aligned} C(z) &= \sum_{n \geq 0} \eta(n) z^{2n} - \frac{z}{2} \sum_{n \geq 0} (\eta(n) + \eta(n+1)) z^{2n} \\ &= C(z^2) - \frac{z}{2} C(z^2) - \frac{1}{2z} (C(z^2) - 1). \end{aligned}$$

Multiplication by $2z$ and simplification give the result. $\square$

This equation places autocorrelation beside E and T in the same Mahler framework, but with a quadratic weight $(1 - z)^2$ that reflects pair products rather than individual signs.

16.3 Finite Riesz products

On the unit circle,

$$\begin{aligned} \frac{1}{2^m} |P_m(e^{2\pi i x})|^2 &= \prod_{j=0}^{m-1} \frac{1}{2} |1 - e^{2\pi i 2^j x}|^2 \\ &= \prod_{j=0}^{m-1} (1 - \cos(2\pi 2^j x)). \end{aligned} \tag{173}$$

Define the probability measures

$$d\mu_m(x) = \prod_{j=0}^{m-1} (1 - \cos(2\pi 2^j x)) dx \quad \text{on } \mathbb{R}/\mathbb{Z}. \tag{174}$$

The integral is one because the constant Fourier coefficient of the product is one.

Theorem 16.3 (Classical spectral limit). *The measures μ_m converge weakly to a probability measure μ whose Fourier coefficients are*

$$\hat{\mu}(k) = \eta(k). \tag{175}$$

The limiting Thue–Morse spectral (or diffraction) measure μ is purely singular continuous: it has neither atoms nor an absolutely continuous component.

This classical result is developed in the substitution and diffraction literature; see, for example, [\[12, 5, 1\]](#). The finite identity [equation \(173\)](#), however, is elementary and exact. It shows that the same polynomial whose coefficients are ε_n also generates the finite spectral intensities.

16.4 A renormalization identity for distribution functions

Let $F(x) = \mu([0, x])$ on $[0, 1]$. Although F has no elementary closed form, the doubling structure of the Riesz products gives renormalization relations for its Fourier coefficients through [equations \(168\) and \(169\)](#). The measure is therefore a natural continuous analogue of a 2-automatic object: its coefficient sequence has a finite recursive description even though the measure itself is singular and non-atomic.

17 Laplace, Mellin, Dirichlet, and finite-product transforms

17.1 Laplace transform of the coefficient sequence

For $t > 0$,

$$\sum_{n \geq 0} \varepsilon_n e^{-nt} = E(e^{-t}) = \prod_{j \geq 0} (1 - e^{-2^j t}). \tag{176}$$

The zero–one version is

$$\sum_{n \geq 0} \tau(n) e^{-nt} = \frac{1}{2(1 - e^{-t})} - \frac{1}{2} \prod_{j \geq 0} (1 - e^{-2^j t}). \quad (177)$$

These are exact transforms, not asymptotic approximations.

17.2 Shifted Dirichlet series

For $a > 0$ and $\Re s > 0$, define

$$D(s, a) = \sum_{n \geq 0} \frac{\varepsilon_n}{(n + a)^s}. \quad (178)$$

The series converges by Dirichlet summation because the prefix sums of ε_n are bounded and $(n + a)^{-s}$ has controlled differences in every closed right half-plane $\Re s > 0$.

Proposition 17.1 (Mellin representation). *For $a > 0$ and $\Re s > 0$,*

$$D(s, a) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-at} E(e^{-t}) dt. \quad (179)$$

Equivalently,

$$D(s, a) = \frac{1}{\Gamma(s)} \int_0^1 (-\log x)^{s-1} x^{a-1} E(x) dx. \quad (180)$$

For $s = 1$ this reduces to

$$D(1, a) = \int_0^1 x^{a-1} E(x) dx. \quad (181)$$

Proof. Insert $(n + a)^{-s} = \Gamma(s)^{-1} \int_0^\infty t^{s-1} e^{-(n+a)t} dt$ and use Abel convergence. The substitution $x = e^{-t}$ gives the second form. $\square$

17.3 A dyadic functional equation for $D(s, a)$

Separating even and odd indices gives

$$D(s, a) = 2^{-s} \left(D\left(s, \frac{a}{2}\right) - D\left(s, \frac{a+1}{2}\right) \right). \quad (182)$$

This is the Dirichlet-series shadow of $\varepsilon_{2n} = \varepsilon_n$ and $\varepsilon_{2n+1} = -\varepsilon_n$. Repeated application produces rapidly convergent binary decompositions for numerical evaluation.

17.4 Positivity of dyadic reciprocal-power blocks

Apply the cubature identity to $f(x) = (x + a)^{-s}$ with $a, s > 0$. Since

$$f^{(m)}(x) = (-1)^m (s)_m (x + a)^{-s-m}, \quad (s)_m = s(s+1) \cdots (s+m-1),$$

we obtain the exact positive formula

$$\sum_{n=0}^{2^m-1} \frac{\varepsilon_n}{(n + a)^s} = 2^{\binom{m}{2}} (s)_m \int_{[0,1]^m} \frac{du}{(a + \sum_{j=0}^{m-1} 2^j u_j)^{s+m}} > 0. \quad (183)$$

Thus every dyadic truncation of the shifted Dirichlet series is positive. The formula is a useful analytic strengthening of Prouhet cancellation.

17.5 Logarithmic block products

For $a > 0$, apply [theorem 11.3](#) to $f(x) = \log(x + a)$. Since for $m \geq 1$

$$f^{(m)}(x) = (-1)^{m-1}(m-1)!(x+a)^{-m},$$

we get

$$\sum_{n=0}^{2^m-1} \varepsilon_n \log(n+a) = -2^{\binom{m}{2}}(m-1)! \int_{[0,1]^m} \frac{d\mathbf{u}}{(a + \sum_{j=0}^{m-1} 2^j u_j)^m} < 0. \quad (184)$$

Exponentiating,

$$0 < \prod_{n=0}^{2^m-1} (n+a)^{\varepsilon_n} < 1. \quad (185)$$

This sign statement is not obvious from the original product but follows immediately from the finite-difference representation.

18 Thue–Morse constants and infinite products

18.1 Base- b Thue–Morse–Mahler numbers

For an integer $b \geq 2$, define

$$\xi_b = \sum_{n \geq 0} \frac{\tau(n)}{b^{n+1}}. \quad (186)$$

From [equation \(86\)](#),

$$\xi_b = \frac{1}{2(b-1)} - \frac{1}{2b} \prod_{j \geq 0} \left(1 - b^{-2^j}\right). \quad (187)$$

For $b = 2$, ξ_2 is the usual Thue–Morse constant $0.0110100110010110 \dots_2$ interpreted as a binary real.

The associated signed number is

$$\chi_b = \sum_{n \geq 0} \frac{\varepsilon_n}{b^{n+1}} = \frac{1}{b} E(1/b), \quad (188)$$

and

$$\xi_b = \frac{1}{2(b-1)} - \frac{1}{2} \chi_b. \quad (189)$$

18.2 Transcendence and irrationality exponent

Mahler’s method applied to the functional equation $E(z) = (1-z)E(z^2)$ proves that the Thue–Morse–Mahler numbers are transcendental; this goes back to Mahler’s 1929 paper [\[11\]](#). A much finer theorem of Bugeaud determines their rational approximation exponent [\[6\]](#).

Theorem 18.1 (Mahler; Bugeaud). *For every integer $b \geq 2$, the Thue–Morse–Mahler number ξ_b is transcendental. Its irrationality exponent is exactly*

$$\mu(\xi_b) = 2. \quad (190)$$

The value 2 is the smallest possible irrationality exponent for an irrational real number. Bugeaud’s proof uses Padé approximants built from nonvanishing Hankel determinants; the determinant input is summarized in [section 20](#).

18.3 A two-parameter Thue–Morse product

For $a, b > 0$, define the conditionally convergent product

$$G(a, b) = \prod_{n \geq 0} \left(\frac{n+a}{n+b} \right)^{\varepsilon_n}, \quad (191)$$

interpreted through logarithms and natural or dyadic truncation. Bounded prefix sums ensure convergence because $\log((n+a)/(n+b)) = \mathcal{O}(n^{-1})$ with first difference $\mathcal{O}(n^{-2})$.

Proposition 18.2 (Even–odd product equation).

$$G(a, b) = G\left(\frac{a}{2}, \frac{b}{2}\right) G\left(\frac{1+b}{2}, \frac{1+a}{2}\right). \quad (192)$$

Also $G(a, a) = 1$ and $G(a, b)G(b, c) = G(a, c)$.

Proof. Split the product into even and odd indices, use $\varepsilon_{2n} = \varepsilon_n$ and $\varepsilon_{2n+1} = -\varepsilon_n$, and cancel the powers of 2 inside each ratio. $\square$

18.4 The Woods–Robbins product

The best-known evaluation is

$$\prod_{n \geq 0} \left(\frac{2n+1}{2n+2} \right)^{\varepsilon_n} = G\left(\frac{1}{2}, 1\right) = \frac{1}{\sqrt{2}}. \quad (193)$$

A short proof uses only even–odd splitting [16, 14]. Put

$$A = \prod_{n \geq 0} \left(\frac{2n+1}{2n+2} \right)^{\varepsilon_n}, \quad B = \prod_{n \geq 1} \left(\frac{2n}{2n+1} \right)^{\varepsilon_n}. \quad (194)$$

Then

$$\begin{aligned} AB &= \frac{1}{2} \prod_{n \geq 1} \left(\frac{n}{n+1} \right)^{\varepsilon_n} \\ &= \frac{1}{2} \prod_{j \geq 0} \left(\frac{2j+1}{2j+2} \right)^{\varepsilon_{2j+1}} \prod_{j \geq 1} \left(\frac{2j}{2j+1} \right)^{\varepsilon_{2j}} \\ &= \frac{1}{2} A^{-1} B. \end{aligned} \quad (195)$$

Cancellation gives $A^2 = 1/2$, and positivity gives the stated value.

18.5 Three dyadic block-product limits

The same product calculus gives

$$\lim_{m \rightarrow \infty} \prod_{k=0}^{2^m-1} \left(k + \frac{1}{2} \right)^{\varepsilon_k} = \frac{1}{2}, \quad (196)$$

$$\lim_{m \rightarrow \infty} \prod_{k=0}^{2^m-1} (k+1)^{\varepsilon_k} = \frac{1}{\sqrt{2}}, \quad (197)$$

$$\lim_{m \rightarrow \infty} \prod_{k=0}^{2^m-1} (k+1)^{(-1)^k \varepsilon_k} = \frac{1}{2\sqrt{2}}. \quad (198)$$

Indeed, pairing adjacent factors in the first two products reduces them to $G(1/4, 3/4)$ and $G(1/2, 1)$, respectively. The first evaluation is a special case of the Thue–Morse product identities of Allouche, Riasat, and Shallit [3], and the second is [equation \(193\)](#). For $m \geq 2$, pairing the third product gives the exact finite relation

$$\prod_{k < 2^m} (k+1)^{(-1)^k \varepsilon_k} = \left(\prod_{n < 2^{m-1}} \left(n + \frac{1}{2} \right)^{\varepsilon_n} \right) \left(\prod_{n < 2^{m-1}} (n+1)^{\varepsilon_n} \right),$$

because $\sum_{n < 2^{m-1}} \varepsilon_n = 0$. Its limit is therefore the product of the first two limits. Dyadic truncation is natural because it respects the exact block recursion.

19 Base- q digit characters and root-of-unity generalization

19.1 The generalized sequence

Let $q \geq 2$, let $s_q(n)$ be the sum of the base- q digits of n , and let $\zeta \neq 1$ be a q th root of unity. Define

$$u_n = \zeta^{s_q(n)}. \quad (199)$$

If $0 \leq r < q$, then

$$u_{qn+r} = \zeta^r u_n. \quad (200)$$

Thus (u_n) is q -automatic and q -multiplicative across digit blocks.

19.2 Finite and infinite products

For $m \geq 0$,

$$\sum_{n=0}^{q^m-1} u_n z^n = \prod_{j=0}^{m-1} \left(1 + \zeta z^{q^j} + \zeta^2 z^{2q^j} + \cdots + \zeta^{q-1} z^{(q-1)q^j} \right). \quad (201)$$

For $|z| < 1$, the infinite generating series

$$U(z) = \sum_{n \geq 0} u_n z^n \quad (202)$$

satisfies

$$U(z) = \left(\sum_{r=0}^{q-1} \zeta^r z^r \right) U(z^q). \quad (203)$$

The binary signed Thue–Morse case is $q = 2$, $\zeta = -1$.

19.3 Generalized Prouhet cancellation

At $z = 1$, every factor in [equation \(201\)](#) vanishes because $1 + \zeta + \cdots + \zeta^{q-1} = 0$. For a nontrivial q th root the zero of each factor is simple. Therefore the full product has a zero of order m at $z = 1$.

Theorem 19.1 (Root-of-unity Prouhet identity). *For $0 \leq r < m$,*

$$\sum_{n=0}^{q^m-1} \zeta^{s_q(n)} n^r = 0. \quad (204)$$

Choose a primitive q th root ω and apply [theorem 19.1](#) successively to $\zeta = \omega^\ell$ for $1 \leq \ell < q$. Finite Fourier inversion then shows that the q residue classes of $s_q(n)$ modulo q have equal sums of powers through degree $m - 1$. This is the q -part Prouhet partition; the familiar even/odd digit-sum partition is its smallest case.

19.4 Multivariate digit weights

More generally, choose weights $w_0, \dots, w_{q-1}$ and define $u_n = \prod_j w_{d_j(n)}$ from the base- q digits $d_j(n)$. Then

$$\sum_{n < q^m} u_n z^n = \prod_{j=0}^{m-1} \left(\sum_{r=0}^{q-1} w_r z^{r q^j} \right). \quad (205)$$

If the digit polynomial $p(z) = \sum_r w_r z^r$ has a zero of order d at $z = 1$, then every factor $p(z^{q^j})$ has the same zero order, so the finite product in [equation \(205\)](#) has a zero of order dm . Consequently its weighted power moments vanish through degree $dm - 1$. This places Thue–Morse in a broad class of q -multiplicative quadrature weights.

20 Hankel determinants, Padé approximation, and natural boundaries

20.1 Hankel matrices

For a sequence $c = (c_n)_{n \geq 0}$, its order- r Hankel determinant is

$$H_r(c) = \det(c_{i+j})_{0 \leq i, j < r}. \quad (206)$$

The Hankel matrix records whether the coefficient sequence admits low-order linear recurrences; nonvanishing determinants guarantee the existence of near-diagonal Padé approximants.

Theorem 20.1 (Allouche–Peyrière–Wen–Wen). *For the signed Thue–Morse sequence $\varepsilon = (\varepsilon_n)_{n \geq 0}$,*

$$H_r(\varepsilon) \neq 0 \quad \text{for every } r \geq 1. \quad (207)$$

Consequently the Padé approximant of type $[r - 1/r]$ to $E(z)$ exists for every r .

This theorem was proved by Allouche, Peyrière, Wen, and Wen [4]; a later combinatorial proof was given by Bugeaud and Han [7]. It is one of the striking places where automatic structure controls a determinant family of unbounded size.

20.2 Connection with irrationality measure

If $H_r(E) \neq 0$, the Padé approximant $A_r(z)/B_r(z)$ satisfies

$$E(z) - \frac{A_r(z)}{B_r(z)} = c_r z^{2r} + \mathcal{O}(z^{2r+1}), \quad c_r \neq 0. \quad (208)$$

Evaluating suitable Mahler iterates at $z = 1/b$ produces rational approximations to $E(1/b)$ and hence to ξ_b . The density of nonvanishing Hankel determinants prevents long gaps in the approximation orders. This is the determinant mechanism behind Bugeaud’s exact exponent $\mu(\xi_b) = 2$.

20.3 Natural boundary

The coefficients of $E(z)$ are integers and its radius of convergence is one. They are not eventually periodic, so E is not rational. The Pólya–Carlson theorem [9] therefore implies:

Theorem 20.2 (Natural boundary). *The unit circle is a natural boundary for $E(z)$. Hence $E(z)$ is transcendental over $\mathbb{C}(z)$.*

The same conclusion holds for $T(z)$ because of [equation \(86\)](#). This sharp contrast is worth emphasizing:

$$\left. \begin{array}{l} \text{over } \mathbb{F}_2(z) \\ \text{over } \mathbb{C}(z) \end{array} \right| \begin{array}{l} \Theta(z) \text{ is quadratic algebraic,} \\ T(z) \text{ is transcendental with a natural boundary.} \end{array} \quad (209)$$

Automaticity is therefore algebraicity in the matching finite characteristic, not in characteristic zero.

21 Combinatorics on words and dynamical structure

21.1 Uniform recurrence and frequencies

The substitution $0 \mapsto 01, 1 \mapsto 10$ is primitive. Standard substitution theory implies that its fixed point is uniformly recurrent and its shift orbit closure is minimal and uniquely ergodic. The substitution matrix

$$M = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \quad (210)$$

has normalized Perron eigenvector $(1/2, 1/2)$, so the frequencies of 0 and 1 are both $1/2$. At dyadic block lengths this equality is exact by [theorem 7.2](#).

21.2 Overlap-freeness

A word is an overlap if it has the form $aXaXa$. Thue’s classical theorem states that the infinite fixed point of [equation \(6\)](#) is overlap-free. In particular it is cube-free. The property is stable under the substitution because any putative shortest overlap in a substituted word can be desubstituted to a shorter overlap, after a finite boundary check. This theorem was the historical starting point for the sequence, long before its modern automatic, spectral, and Diophantine roles.

21.3 Aperiodicity

The reflection rule gives a short contradiction to periodicity. Suppose that $p \geq 1$ is a period and choose any m with $2^m > p$. Periodicity and [equation \(74\)](#) imply

$$-1 = \varepsilon_{2^m} = \varepsilon_{2^m-p} = \varepsilon_{(2^m-1)-(p-1)} = (-1)^m \varepsilon_{p-1}.$$

This identity would have to hold for arbitrarily large even and odd m , which is impossible. Hence the sequence is not periodic. It is not eventually periodic either: if period p worked beyond an index N_0 , then for any fixed $r \geq 0$ one could choose m so large that $2^m > r + p$ and $2^m + r \geq N_0$. Periodicity inside the complemented dyadic block would give

$$-\varepsilon_{r+p} = \varepsilon_{2^m+r+p} = \varepsilon_{2^m+r} = -\varepsilon_r,$$

so p would actually be a global period, a contradiction. Aperiodicity is also implied by overlap-freeness.

21.4 Factor complexity and automaticity

The number of distinct length- n factors grows linearly rather than exponentially. Exact piecewise formulae are known, but the important structural point here is that a primitive constant-length substitution has low factor complexity, zero topological entropy, and finite-state self-similarity. These features coexist with a singular continuous spectral measure, showing that low combinatorial complexity need not imply pure point spectrum.

22 Worked examples and computational checks

22.1 A sixteen-term comparison table

Let $C_n = \binom{2n}{n}$ and $\mathcal{O}(n)$ be the number of odd entries in Pascal row n .

n	binary	$s_2(n)$	$\tau(n)$	ε_n	$v_2(C_n)$	$\mathcal{O}(n)$
0	0000	0	0	1	0	1
1	0001	1	1	-1	1	2
2	0010	1	1	-1	1	2
3	0011	2	0	1	2	4
4	0100	1	1	-1	1	2
5	0101	2	0	1	2	4
6	0110	2	0	1	2	4
7	0111	3	1	-1	3	8
8	1000	1	1	-1	1	2
9	1001	2	0	1	2	4
10	1010	2	0	1	2	4
11	1011	3	1	-1	3	8
12	1100	2	0	1	2	4
13	1101	3	1	-1	3	8
14	1110	3	1	-1	3	8
15	1111	4	0	1	4	16

Every numeric column is determined by $s_2(n)$, but each displays a different mathematical encoding.

22.2 Example: the product P_4

The fourth product is

$$\begin{aligned}
 P_4(z) &= (1-z)(1-z^2)(1-z^4)(1-z^8) \\
 &= 1 - z - z^2 + z^3 - z^4 + z^5 + z^6 - z^7 \\
 &\quad - z^8 + z^9 + z^{10} - z^{11} + z^{12} - z^{13} - z^{14} + z^{15}.
 \end{aligned}$$

At $z = 1$ it has a zero of order four. Therefore the coefficients have equal moments through degree three. The first nonzero moment is

$$\sum_{n=0}^{15} \varepsilon_n n^4 = 4!2^6 = 1536.$$

At a primitive cubic root ω , $P_4(\omega) = 9$, and the modulo-three rarefied vector is $(6, -3, -3)$.

22.3 Example: sparse support at $N = 13$

Since $13 = (1101)_2$, $J(13) = \{0, 2, 3\}$ and $s_2(13) = 3$. Hence

$$\begin{aligned}\sum_{k \prec_2 13} \varepsilon_k z^k &= (1 - z)(1 - z^4)(1 - z^8) \\ &= 1 - z - z^4 + z^5 - z^8 + z^9 + z^{12} - z^{13}.\end{aligned}$$

The signed submask moments of degree 0, 1, 2 vanish, and

$$\sum_{k \prec_2 13} \varepsilon_k k^3 = (-1)^3 3! 2^{0+2+3} = -192.$$

22.4 Example: carries

For $a = 23 = (10111)_2$ and $b = 9 = (01001)_2$,

$$a + b = 32 = (100000)_2, \quad s_2(a) + s_2(b) - s_2(a + b) = 4 + 2 - 1 = 5.$$

Thus five carries occur and

$$\varepsilon_{32} = \varepsilon_{23} \varepsilon_9 (-1)^5 = (-1)(+1)(-1) = +1.$$

22.5 Reference algorithms

The direct bit-count algorithm is the fastest practical evaluator when a machine popcount is available.

Listing 1: Reference checks for the principal identities

```
from math import comb

def tau(n: int) -> int:
    if n < 0:
        raise ValueError("n must be nonnegative")
    return n.bit_count() & 1

def epsilon(n: int) -> int:
    return 1 - 2 * tau(n)

def v2(x: int) -> int:
    if x <= 0:
        raise ValueError("x must be positive")
    return (x & -x).bit_length() - 1

def epsilon_central_binomial(n: int) -> int:
    return -1 if (v2(comb(2*n, n)) & 1) else 1

def rarefied_mod3(m: int) -> tuple[int, int, int]:
    out = [0, 0, 0]
    for n in range(1 << m):
        out[n % 3] += epsilon(n)
    return tuple(out)

for n in range(1000):
    assert epsilon(n) == epsilon_central_binomial(n)

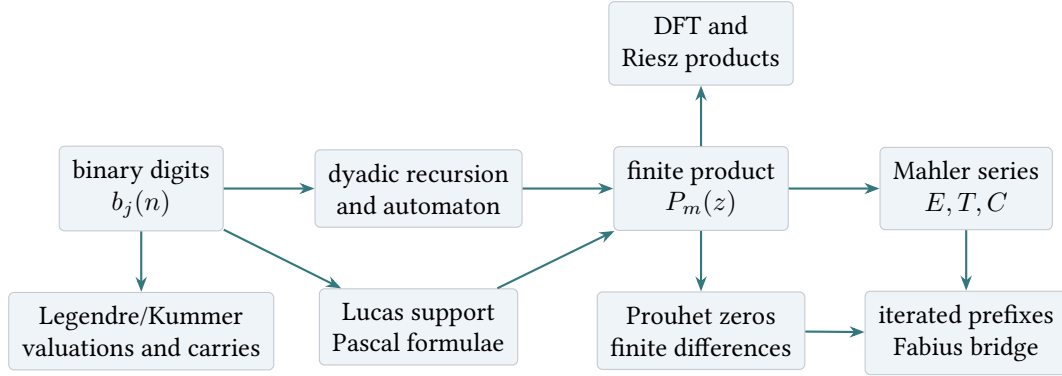
for m in range(1, 12):
```

```

q = m // 2
expected = ((2 * 3**(q-1), -3**(q-1), -3**(q-1))
            if m % 2 == 0
            else (3**q, -3**q, 0))
assert rarefied_mod3(m) == expected

```

22.6 Dependency map of the formulae



23 Formalization roadmap for ProveIt

23.1 Already central in the existing development

The repository already contains Lean developments around the binomial–logarithm formula, finite generating products, prefix sums, and the Fabius bridge. The consolidated article suggests organizing further work by mathematical dependency rather than by the order in which formulae were discovered.

23.2 Layer 1: binary and finite algebra

A foundational module should contain:

- the recursions $\varepsilon_{2n} = \varepsilon_n$ and $\varepsilon_{2n+1} = -\varepsilon_n$;
- block multiplicativity $\varepsilon_{2^m a + b} = \varepsilon_a \varepsilon_b$;
- the XOR character law;
- the finite coefficient product $P_m(z)$ over a general commutative ring;
- reciprocity and the cyclotomic factorization of P_m .

These statements need only finite sums, finite products, and bit lemmas.

23.3 Layer 2: valuations and Pascal support

The next layer depends on base-two digit-sum identities:

- Legendre’s formula $v_2(n!) = n - s_2(n)$;
- $v_2\binom{2n}{n} = s_2(n)$;
- Kummer’s carry formula and the carry-corrected sign law;
- Lucas’s submask criterion and the Glaisher count;
- the modulo-three log-free formula;

- odd multinomial support $r^{s_2(n)}$.

For formalization, it is preferable to state equalities in $\mathbb{N}$ before taking parity in $\mathbb{Z}/2\mathbb{Z}$.

23.4 Layer 3: polynomials, moments, and finite differences

This layer should prove:

- the order- m zero of P_m at 1;
- Prouhet cancellation for powers and falling factorials;
- the operator identity with mixed finite differences;
- the exact cubature formula by iterated fundamental theorem of calculus;
- the all-moment composition formula;
- sparse submask Prouhet identities.

A reusable abstraction would take arbitrary steps $h_0, \dots, h_{m-1}$ and weights from $\prod_j (1 - z^{h_j})$, with Thue–Morse as the dyadic instance.

23.5 Layer 4: formal power series

The formal series layer should include:

- $E(z) = (1 - z)E(z^2)$ and the coefficient recurrence from its logarithmic derivative;
- the Bell-polynomial and Hessenberg determinant formulae;
- the characteristic-two quadratic equation for Θ ;
- the integer algebraic lift and its coefficientwise parity theorem;
- the iterated-prefix generating series $E(z)/(1 - z)^k$;
- the autocorrelation functional equation, once existence and recursion of η are available.

The finite-product versions should be proved first; analytic convergence can then be added as a separate wrapper.

23.6 Layer 5: Fourier and analysis

The final layer contains:

- root-of-unity evaluation and the exact DFT sine product;
- the modulo-three rarefaction theorem;
- normalized finite Riesz products and their Fourier coefficients;
- Mellin and Dirichlet integral representations;
- finite reciprocal-power positivity and logarithmic-product inequalities;
- the Fabius/up-function limits already developed elsewhere in the repository.

Gelfond’s sharp estimate, singular continuity, transcendence, and irrationality measure should initially enter as cited external theorems rather than targets for immediate re-formalization.

23.7 Suggested module graph

Listing 2: Possible Lean module decomposition

```
ThueMorse.Basic
-> ThueMorse.Blocks
-> ThueMorse.BooleanCharacter
-> ThueMorse.FiniteProduct

ThueMorse.DigitSum
-> ThueMorse.Valuation
-> ThueMorse.Carries
-> ThueMorse.Pascal
-> ThueMorse.Multinomial

ThueMorse.FiniteProduct
-> ThueMorse.Prouhet
-> ThueMorse.FiniteDifference
-> ThueMorse.Moments
-> ThueMorse.SparseProuhet
-> ThueMorse.DFT
-> ThueMorse.RarefactionMod3

ThueMorse.FormalSeries
-> ThueMorse.Mahler
-> ThueMorse.RulerRecurrence
-> ThueMorse.BellDeterminant
-> ThueMorse.CharTwo
-> ThueMorse.IntegerLift
-> ThueMorse.IteratedPrefix

ThueMorse.Autocorrelation
-> ThueMorse.RieszProduct
-> ThueMorse.AutocorrelationGenerating
```

24 Conclusions

The four drafts become considerably clearer when organized around one finite product. Binary digits give its coefficient signs; Lucas turns the same binary support into Pascal’s odd positions; Kummer measures the carry obstruction to multiplicativity; the zero at 1 gives Prouhet cancellation and exact finite differences; unit-circle evaluation gives ordinary Fourier transforms; squared moduli give Riesz products; division by powers of $1 - z$ gives iterated prefix sums; and passage to the infinite product gives Mahler equations and analytic constants.

The most useful new consequences of the consolidation are not isolated formulae but bridges. The modulo-three rarefied sums show that cubic-root Fourier extremizers have exact arithmetic-progression meaning. The functional equation for the autocorrelation series shows that the spectral coefficients belong to the same Mahler universe as the original sequence. The characteristic-two quadratic and the characteristic-zero natural boundary show how strongly the ambient field changes the algebraic nature of an automatic series. Finally, the sparse Pascal-support identities reveal that every binary row index carries its own Prouhet system, with cancellation order exactly equal to its digit sum.

The resulting atlas is therefore more than a catalogue. It is a dependency network in which the sequence’s arithmetic, combinatorial, harmonic, and analytic appearances can be translated into one another by short exact arguments.

A The first sixty-four values

The zero–one word, grouped into blocks of sixteen, is

$$\begin{aligned} 0-15 : & \ 0110100110010110, \\ 16-31 : & \ 1001011001101001, \\ 32-47 : & \ 1001011001101001, \\ 48-63 : & \ 0110100110010110. \end{aligned}$$

The corresponding signed blocks are

$$\begin{aligned} 0-15 : & \ + - - + - + + - - + + - + - - +, \\ 16-31 : & \ - + + - + - - + + - - + - + + -, \\ 32-47 : & \ - + + - + - - + + - - + - + + -, \\ 48-63 : & \ + - - + - + + - - + + - + - - +. \end{aligned}$$

The repeated middle blocks and reversed signs at dyadic boundaries are direct instances of [equations \(7\)](#) and [\(74\)](#).

B Identity crosswalk

Target	Exact formula	Main bridge
Individual sign	$\varepsilon_n = (-1)^{s_2(n)}$	definition
Digit floors	$s_2(n) = n - \sum_{j \geq 1} \lfloor n/2^j \rfloor$	telescoping bit extraction
Factorial valuation	$\varepsilon_n = (-1)^{n-v_2(n!)}$	Legendre
Central binomial	$\varepsilon_n = (-1)^{v_2\binom{2n}{n}}$	Legendre plus shift invariance
Carries	$\varepsilon_{a+b} = \varepsilon_a \varepsilon_b (-1)^{v_2\binom{a+b}{a}}$	Kummer
Pascal support	$\mathcal{O}(n) = 2^{s_2(n)}$	Lucas
Log-free Pascal	$\tau(n) = [\mathcal{O}(n)]_3 - 1$	powers of two modulo three
Multinomial support	$\mathcal{O}_r(n) = r^{s_2(n)}$	carry-free digit allocation
Finite product	$P_m(z) = \prod_{j < m} (1 - z^{2^j})$	binary subset expansion
Prouhet	$\sum_{n < 2^m} \varepsilon_n n^r = 0 \text{ for } r < m$	order- m zero at 1
Finite differences	$\sum \varepsilon_n f(x+n) = (-1)^m \prod_j \Delta_{2^j} f(x)$	shift-operator factorization
DFT	$\widehat{\varepsilon}_m(k) = P_m(e^{-2\pi i k/2^m})$	evaluation at roots of unity
Rarefaction mod 3	equations (163) and (164)	cubic-root filter
Riesz product	$2^{-m} P_m(e^{2\pi i x}) ^2 = \prod_j (1 - \cos 2\pi 2^j x)$	squared Fourier magnitude
Mahler series	$E(z) = (1-z)E(z^2)$	least-significant-bit split
Autocorrelation series	$2zC(z) = 1 - (1-z)^2 C(z^2)$	even–odd correlation recursion
Characteristic two	$(1+z)^3 \Theta^2 + (1+z)^2 \Theta + z = 0$	Frobenius $\Theta(z^2) = \Theta(z)^2$
Iterated prefixes	$\sum S_n^{(k)} z^n = E(z)/(1-z)^k$	discrete integration

C Source-draft consolidation map

Draft	Distinct emphasis retained here	Principal consolidated sections
Draft 1	central-binomial valuation, carries, floor and Boolean formulae, logarithmic derivative, Bell and determinant forms, analytic transforms	Sections 2–4, 9–10, 18
Draft 2	master catalogue, higher dyadic moments, exact cubature, automata, tensor/Walsh forms, DFT and computational organization	Sections 2, 11–12, 15, 23
Draft 3	fractional-part formula, Catalan and multinomial support, sparse Pascal Prouhet systems, characteristic-two algebraicity, autocorrelation and Riesz products, base- q extension	Sections 3–6, 12, 16–17, 21
Draft 4	sinc sign encoding, evil/odious enumerators, log-free Pascal formula, one-parameter row statistic, integer algebraic lift, infinite products, and iterated-prefix/Fabius bridge	Sections 3, 7–8, 10, 13, 19–20
New consolidation	exact modulo-three rarefied sums, autocorrelation generating equation, Gelfond-extremizer synthesis, natural-boundary contrast, integrated formalization graph	Sections 14, 16, 22, 24

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